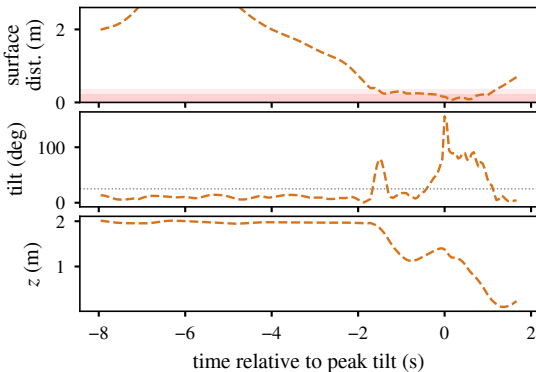
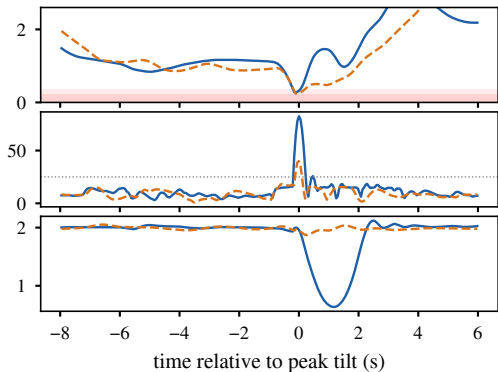


(a) contact leading to loss



(b) near-contact with recovery



-- rect, PID-LQR, iris_2
 — plus, PID- H_∞ , iris_2
 -- plus, PID-LQR, iris_2
 body contact
 rotor contact possible